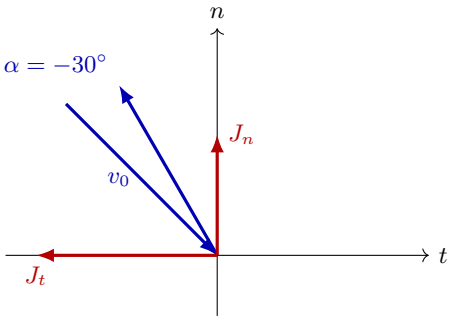
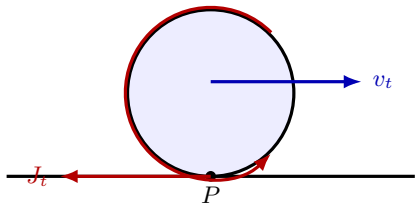


$v_t' = v_t - J_t/m < 0$

$\omega > 0$：逆时针



$u_P = v_t + R\omega$

$\Delta K = -J_t u_P + \frac{1}{2} J_t^2 (1/m + R^2/I)$